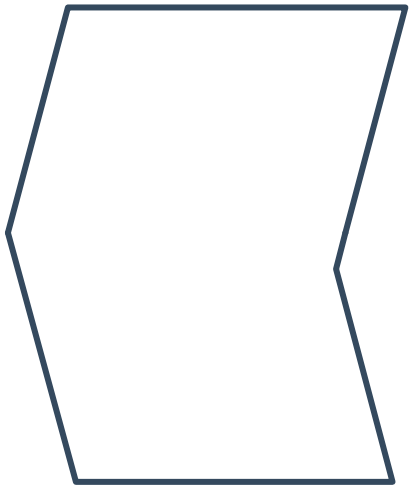
Interior angles of polygons

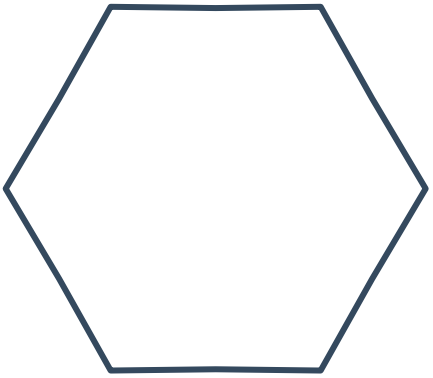
1 Consider the non-convex hexagon:



a What is the least number of triangles that it can be divided into?

b What is the interior angle sum of the non-convex hexagon?

2 Consider the convex hexagon:

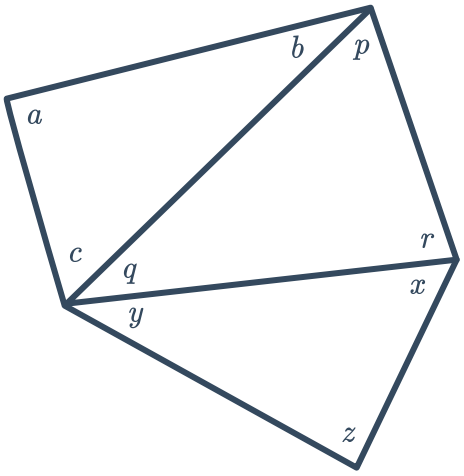


a Find the least number of triangles that the hexagon can be divided into.

b Find the interior angle sum of the hexagon.

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3 Consider the pentagon:



a Find the value of the following:

i $a+b+c$ **ii** $p+q+r$ **iii** $x+y+z$

b Deduce the angle sum of the pentagon.

c How many non-overlapping triangles can an n-sided figure be split into?

d Determine whether each of the following is the rule for the angle sum of an n-sided polygon:

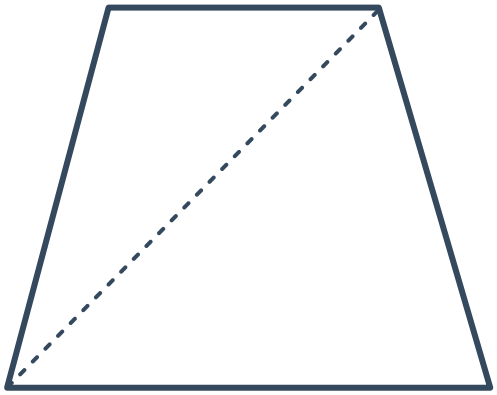
i $n \times 180$

ii $(n+2) \times 180$

iii $(n-2) \times 180$

iv $(n-2) \times 360$

4 Consider the figure. The quadrilateral is divided into two triangles (this is the least number of triangles that a quadrilateral can be divided into).



The interior angle sum of a quadrilateral is therefore equal to double the interior angle sum of a triangle.

Complete the following table. 'Least Number of Triangles' means the least number of triangles that the shape can be divided up into.

	Number Of Sides	Least Number of Triangles	Interior angle sum
Pentagon			
Hexagon			
Heptagon			
Octagon			
Nonagon			
Decagon			

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5 Find the sum of the interior angles in the following figures:

a Quadrilateral

b Twelve-sided polygon

c Fourteen-sided polygon

d Sixteen-sided polygon

e Twenty-sided polygon

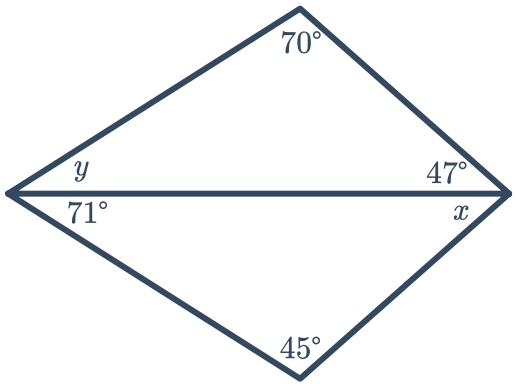
f Forty-sided polygon

6 For each of the following figures:

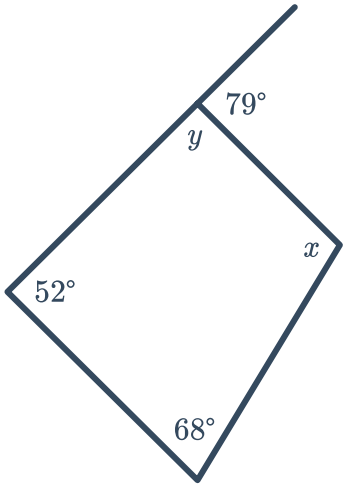
i Find the value of y.

ii Find the value of x.

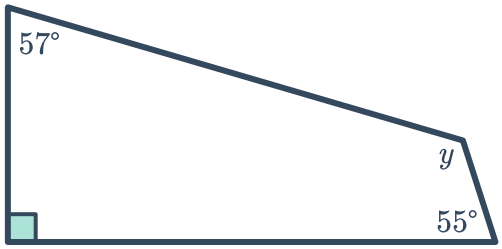
a



b



7 Find the value of y in the following figure:



8 Consider an octagon.

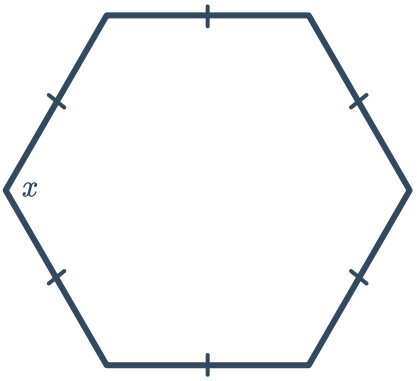
a What is the sum of its interior angles?

b Given that it is regular, find the size of each interior angle.

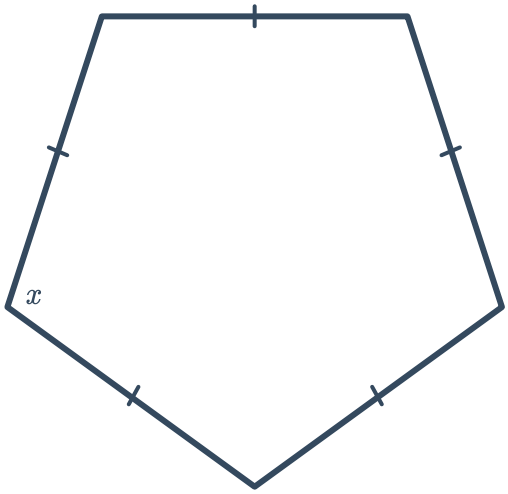
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9 For each of the following regular polygons, find the value of x:

a



b



10 For each of the following polygons:

i Find n, the number of sides.

ii Classify the polygon.

a A polygon with equal internal angles of 120°.

b A polygon with equal internal angles of 135°.

c A polygon with equal internal angles of 60°.

11 Explain why a regular polygon with equal internal angles of 50° is not possible.

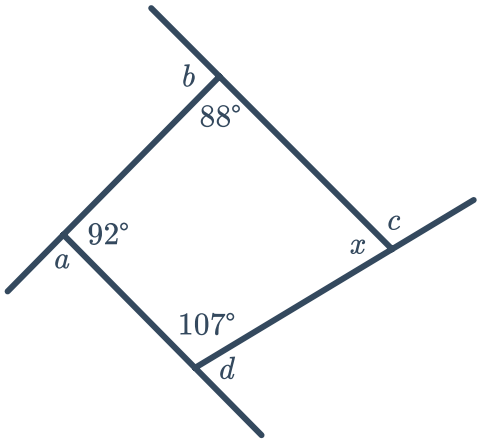
12 Find the interior angle sum of a regular polygon whose exterior angles measure 40°.

13 The sum of the interior angles of a regular 8-sided shape is 1080°. Find the size of each interior angle.

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Exterior angles of polygons

14 Consider the given quadrilateral:



a Find the value of:

i x **ii** a

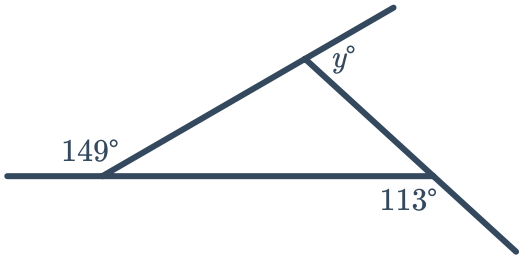
iii b **iv** c

v d **vi** $a+b+c+d$

b What property about quadrilaterals has been shown?

15 Consider the following figure:

Find the value of y.

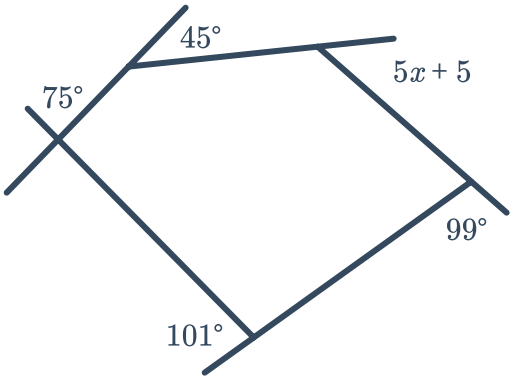


16 Find the value of x:

a



b



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17 Find the size of one exterior angle for the following regular polygons:

a Pentagon **b** Octagon **c** Triangle **d** Nonagon

18 Find the size of one interior angle for the following regular polygons:

a Hexagon **b** Quadrilateral

c Decagon **d** Twelve-sided polygon

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